

Recurrence-Free Dyadic Formulae for the Fabius and Rvachev Functions

**Explicit partitions, positive compositions, binary telescopes,
integer determinants, and parity-reduced spline extraction**

A companion research article prepared for Vladimir Reshetnikov

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Abstract

We give several completely finite descriptions of the exact rational values of $F(a/2^n)$ and $\text{up}(a/2^n)$. Every auxiliary coefficient is specified by an explicit finite sum or product; no moment recurrence or recursive evaluation of another function value is required. The classical Thue–Morse formula is completed in two ways: a Bernoulli partition expansion, with the Bernoulli numbers themselves expanded into finite power sums, and a positive ordered-composition expansion. A binary-digit telescope removes the long Thue–Morse prefix, while an integer bordered determinant gives a compact arithmetic certificate and a common denominator.

The main moment-free construction uses centered finite uniform splines. At a dyadic argument of depth n , their cutoff dependence is an *exact* polynomial in 4^{-N} of degree at most $\lfloor n/2 \rfloor$, not merely an asymptotic expansion. Lagrange evaluation at zero therefore recovers the function from only $\lfloor n/2 \rfloor + 1$ consecutive spline levels. Substitution of the finite box-volume formula produces a double finite sum involving only integers, factorials, Thue–Morse signs, and q -Pochhammer symbols at $q = 1/4$. We prove the cutoff-polynomial identity, its exact degree at reduced interior dyadics, the connection with the existing half-base formula, and analogous formulae for the Rvachev bump, derivatives, and iterated primitives. The same construction extends to normalized reciprocal-integer-base uniform series. Exact rational tests compare five formulae on 520 dyadic grid cases, together with separate spline, bump, shift, and other-base tests. The accompanying standard-library Python program reproduces these checks.

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1 Scope, conventions, and the formula map

1.1 What is meant here by a closed form

A finite expression is called *recurrence-free* if its evaluation uses only explicit finite index sets, finite sums and products, rational arithmetic, integer powers, factorials, and directly specified combinatorial quantities. Thus a formula involving moments c_j is not yet complete until the c_j have been replaced by finite expressions. A coefficient extractor is useful in a proof, but it is not left as an unexplained operation in the final evaluation formulae. A determinant is permissible because we give its entries explicitly and recall its finite permutation expansion.

Throughout, an empty sum is zero, an empty product is one, and $0^0 = 1$ in polynomial formulae. All displayed index bounds are finite. We write

$$\varepsilon_k = (-1)^{s_2(k)}, \quad T_n = \binom{n+1}{2}, \quad (1)$$

where $s_2(k)$ is the number of ones in the binary expansion of k . This is a direct digit definition, not a recurrence. Equivalently, for any L with $k < 2^L$,

$$\varepsilon_k = \prod_{j=0}^{L-1} (1 - 2(\lfloor k/2^j \rfloor - 2\lfloor k/2^{j+1} \rfloor)).$$

For $0 < q < 1$ we use the finite products

$$(q; q)_r = \prod_{j=1}^r (1 - q^j), \quad \left[\begin{matrix} d \\ j \end{matrix} \right]_q = \frac{(q; q)_d}{(q; q)_j (q; q)_{d-j}}. \quad (2)$$

No infinite q -product is needed to evaluate any dyadic value in this article.

1.2 The three functions must be kept separate

Let $U_1, U_2, \dots$ be independent uniform random variables on $[0, 1]$, and put

$$X = \sum_{j=1}^{\infty} 2^{-j} U_j, \quad F(x) = \mathbb{P}(X \leq x), \quad Y = 2X - 1. \quad (3)$$

The bounded CDF F is zero on $(-\infty, 0]$ and one on $[1, \infty)$. The density of Y is the even bump up, normalized by $\text{up}(0) = 1$ and $\text{supp up} = [-1, 1]$. In particular,

$$\text{up}(t) = F(1 - |t|) \quad (|t| \leq 1), \quad F'(x) = 2 \text{up}(2x - 1). \quad (4)$$

The signed extension is denoted by $\mathcal{F}$, not by F :

$$\mathcal{F}(x) = \sum_{m \geq 0} \varepsilon_m \text{up}(x - 2m - 1). \quad (5)$$

This sum is locally finite, and at most one summand is nonzero. On $[0, 1]$ it equals F , but on the rest of the real line it is not the bounded CDF.

Consequently, a formula for F on the unit interval already answers the question for every bump value:

$$\text{up}(a/2^n) = \begin{cases} F((2^n - |a|)/2^n), & |a| \leq 2^n, \\ 0, & |a| > 2^n. \end{cases} \quad (6)$$

For an arbitrary nonnegative dyadic x , set $m = \lfloor x/2 \rfloor$ and $t = x - 2m$. Then

$$\mathcal{F}(x) = \varepsilon_m F(\min(t, 2 - t)), \quad 0 \leq t < 2. \quad (7)$$

Negative arguments give $\mathcal{F}(x) = 0$. These reductions avoid unnecessarily long prefix sums outside the unit interval.

1.3 Relation to the repository and earlier work

The comparison source is the linked primary exposition in the ProveIt repository, read at commit 2c6baf6738a5fbf3981a27141950edbe5952b03c; its main TeX blob is c5982d732a3caa769547e079efce37d25fb84a4e [1]. Its sections on reciprocal powers and arbitrary dyadics already contain moment recurrences, a Thue–Morse expression, and exact path/composition descriptions of $F(2^{-n})$. The archived companion *The Dyadic Web of the Fabius Function* already connects the half-base formula, binary reduction, moment correction, and discrete extrapolation [2].

We therefore do not present the existence of composition formulae, the Thue–Morse identity, or half-base interpolation as new discoveries. The purpose here is to finish every coefficient into a finite expression, organize genuinely different evaluation routes, and develop the centered, parity-reduced cutoff identity with a precise level bound. Historical priority for every consequence is not asserted. The classical analytic and arithmetic background is developed by Arias de Reyna [3, 4]; Haugland gives another exact dyadic evaluator [5]. The half-base expression posed by Reshetnikov is discussed explicitly in section 9 [6].

Family	Finite ingredients and characteristic feature	Main result
Bernoulli partitions	Multiplicity vectors and finite Bernoulli power sums; no recursive moments	theorem 4.2
Positive compositions	Ordered positive parts and Mersenne factors; coefficient terms are positive	theorem 4.4
Binary telescope	Only the nonzero binary digits of the argument; explicit moment coefficients	theorem 5.2
Integer determinant	A matrix of order $\lfloor n/2 \rfloor + 1$; integer numerator certificate	theorem 6.1
Quarter-base extraction	$\lfloor n/2 \rfloor + 1$ centered spline levels; no moments at all	theorem 8.2
Direct bump extraction	The same weights, applied to finite densities instead of CDFs	theorem 10.1

Table 1: Alternative presentations. All rows are exact and recurrence-free.

2 Analytic facts needed for exact arithmetic

2.1 Smoothness, reflection, and the bump identity

The series in (3) converges absolutely for every outcome, because $0 \leq U_j \leq 1$ and $\sum 2^{-j} = 1$. Replacing every U_j by $1 - U_j$ gives $X \stackrel{d}{=} 1 - X$. Its law has no atoms: conditioning on all but U_1 leaves a translated uniform distribution. Hence

$$F(1 - x) = 1 - F(x) \quad (0 \leq x \leq 1). \quad (8)$$

The characteristic function of the centered variable is

$$\mathbb{E}e^{itY} = \prod_{j \geq 1} \frac{\sin(t/2^j)}{t/2^j}.$$

For any fixed K , the first K factors give a bound $C_K(1 + |t|)^{-K}$; the remaining factors have absolute value at most one. Choosing K arbitrarily large makes the inverse Fourier integral and all its derivatives absolutely convergent. Thus Y has a C^∞ density up, supported in $[-1, 1]$, and F is smooth as well.

The self-similarity $Y \stackrel{d}{=} (V + Y')/2$, with V uniform on $[-1, 1]$ and Y' an independent copy of Y , gives

$$\text{up}(t) = \mathbb{P}(2t - 1 \leq Y' \leq 2t + 1) = F(t + 1) - F(t).$$

On $[-1, 0]$ this is $F(t + 1)$; on $[0, 1]$ it is $1 - F(t) = F(1 - t)$. This proves (4). The identity between the two densities under $Y = 2X - 1$ gives $F'(x) = 2 \operatorname{up}(2x - 1)$. Differentiating the preceding formula also gives

$$\operatorname{up}'(t) = 2 \operatorname{up}(2t + 1) - 2 \operatorname{up}(2t - 1). \quad (9)$$

Smoothness and compact support imply that all positive-order derivatives of F vanish at 0 and 1, and all derivatives of up vanish at -1 and 1.

2.2 Derivative termination at dyadic arguments

Lemma 2.1 (Signed extension and dyadic jets). *The function (5) is smooth and satisfies*

$$\mathcal{F}'(x) = 2\mathcal{F}(2x), \quad \mathcal{F}^{(r)}(x) = 2^{r(r+1)/2} \mathcal{F}(2^r x). \quad (10)$$

At nonnegative integers,

$$\mathcal{F}(2m) = 0, \quad \mathcal{F}(2m + 1) = \varepsilon_m. \quad (11)$$

If $0 < a < 2^n$, then

$$F^{(r)}(a/2^n) = 0 \quad (r > n). \quad (12)$$

Likewise, if $|a| < 2^n$, then

$$\operatorname{up}^{(r)}(a/2^n) = 0 \quad (r > n). \quad (13)$$

Proof. Flatness at the endpoints of each translated bump makes the locally finite sum smooth, including the joining points. By (9), its derivative is

$$2 \sum_{m \geq 0} \varepsilon_m \{ \operatorname{up}(2x - 4m - 1) - \operatorname{up}(2x - 4m - 3) \} = 2\mathcal{F}(2x),$$

because the direct digit definition gives $\varepsilon_{2m} = \varepsilon_m$ and $\varepsilon_{2m+1} = -\varepsilon_m$. Repeated differentiation proves (10); evaluation at bump endpoints and centers gives (11). For an interior unit-interval argument, F and $\mathcal{F}$ agree on a neighborhood. When $r > n$, the number $2^r a/2^n$ is an even integer, so its signed value vanishes. On $(-1, 1)$, $\operatorname{up}(x) = \mathcal{F}(x + 1)$; the same reasoning applies to $2^r(a/2^n + 1)$. At $a = 0, n = 0$ this also proves that every positive-order bump derivative vanishes at zero. $\square$

Remark 2.2 (A finite Taylor jet is not a local polynomial identity). The vanishing in (12) does not assert that F equals its Taylor polynomial near the dyadic point. The finite-spline construction below uses a polynomial identity for a *finite convolution*, together with an exact smoothing identity. It does not replace F by a convergent Taylor series. This distinction is essential to the proof.

3 The master dyadic formula, proved from a finite cube

3.1 A box-volume identity

Lemma 3.1 (Finite weighted-uniform CDF). *For positive $w_1, \dots, w_N$, let $Z = \sum_{j=1}^N w_j U_j$. Then*

$$\mathbb{P}(Z \leq x) = \frac{1}{N! \prod_j w_j} \sum_{\delta \in \{0,1\}^N} (-1)^{|\delta|} \left(x - \sum_{j=1}^N w_j \delta_j \right)_+^N. \quad (14)$$

Here $s_+ = \max(s, 0)$ and $N \geq 1$.

Proof. Changing variables $v_j = w_j U_j$ makes the desired probability the volume of $\{0 \leq v_j \leq w_j, \sum v_j \leq x\}$ divided by $\prod_j w_j$. The unrestricted nonnegative simplex has volume $x_+^N/N!$, as follows by N repeated integrations. Exclude the events $v_j > w_j$ by finite inclusion–exclusion. For a selected set of excluded constraints, translate the corresponding coordinates by w_j ; the resulting simplex has threshold $x - \sum w_j \delta_j$. Boundaries have measure zero. This gives (14). $\square$

Set

$$X_N = \sum_{j=1}^N 2^{-j} U_j, \quad c_j = \mathbb{E} Y^{2j}, \quad C_j = \frac{c_j}{(2j)!}, \quad A_m = \frac{\mathbb{E} X^m}{m!}. \quad (15)$$

In particular, $C_0 = A_0 = 1$. The symbols C_j, A_m will shortly be replaced by explicit sums.

Theorem 3.2 (Master dyadic coefficient). *For $n \geq 1$ and $0 \leq a \leq 2^n$,*

$$F(a/2^n) = 2^{-\binom{n}{2}} \sum_{h=0}^{a-1} \varepsilon_h \sum_{m=0}^n A_m \frac{(a-h-1)^{n-m}}{(n-m)!}, \quad (16)$$

$$= 2^{-T_n} \sum_{h=0}^{a-1} \varepsilon_h \sum_{j=0}^{\lfloor n/2 \rfloor} C_j \frac{(2a-2h-1)^{n-2j}}{(n-2j)!}. \quad (17)$$

Equivalently,

$$F(a/2^n) = \frac{2^{-T_n}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} c_j (2a-2h-1)^{n-2j}. \quad (18)$$

For $n = 0$, the same central expression gives the two allowed values $F(0) = 0$ and $F(1) = 1$.

Proof. Split the infinite series into independent head and tail: $X \stackrel{d}{=} X_n + 2^{-n} X'$. In (14), the weights 2^{-j} have product $2^{-n(n+1)/2}$, and their subset sums are $h/2^n$ for $0 \leq h < 2^n$, with signs ε_h . Conditioning on X' therefore gives

$$F(a/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{2^n-1} \varepsilon_h \mathbb{E}(a-h-X')_+^n.$$

If $h \geq a$, the summand is zero. If $h < a$, then $a-h \geq 1$ and $0 \leq X' \leq 1$, so the positive part can be removed. Reflection of X' changes the expectation into $\mathbb{E}(a-h-1+X)^n$. Expanding the ordinary binomial yields (16). Finally,

$$a-h-1+X = \frac{1}{2}(2a-2h-1+Y),$$

and odd moments of Y vanish. The extra factor 2^{-n} converts $2^{-\binom{n}{2}}$ into 2^{-T_n} , proving the remaining formulae. $\square$

This proof explains exactly why the tail produces only finitely many moments: the dyadic threshold falls between consecutive head knots, so the positive part becomes an ordinary polynomial throughout the whole tail support. Rationality is not assumed anywhere in this argument.

4 Removing all moment recurrences

4.1 Finite Bernoulli numbers and cumulants

The standard Bernoulli convention is $B_1 = -1/2$ [7]. To avoid hiding a recurrence even in this classical sequence, define

$$B_m = \sum_{k=0}^m \frac{1}{k+1} \sum_{v=0}^k (-1)^v \binom{k}{v} v^m. \quad (19)$$

This finite expression agrees with the generating function

$$\frac{z}{e^z - 1} = \sum_{m \geq 0} B_m \frac{z^m}{m!}. \quad (20)$$

Indeed, the exponential generating function of the inner sum is $(1 - e^z)^k$. Summing $u^k/(k+1) = -\log(1-u)/u$ with $u = 1 - e^z$ gives $z/(e^z - 1)$. Coefficientwise, only $k \leq m$ can contribute, since $(1 - e^z)^k$ has order k . Thus the argument is also valid as a formal power-series identity, without an infinite numerical evaluation.

The moment generating functions are

$$M_Y(z) = \mathbb{E}e^{zY} = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}, \quad M_X(z) = e^{z/2} M_Y(z/2). \quad (21)$$

They are entire because the underlying variables are bounded. The logarithm is needed only in a neighborhood of zero. Define the explicit rational numbers

$$\lambda_r = \frac{4^r B_{2r}}{2r(2r)!(4^r - 1)}, \quad \beta_r = \frac{B_{2r}}{2r(2r)!(4^r - 1)}. \quad (22)$$

Then

$$\log M_Y(z) = \sum_{r \geq 1} \lambda_r z^{2r}, \quad \log M_X(z) = \frac{z}{2} + \sum_{r \geq 1} \beta_r z^{2r}. \quad (23)$$

To verify the constants, differentiate $\log(\sinh z/z)$:

$$\coth z - \frac{1}{z} = \sum_{r \geq 1} \frac{2^{2r} B_{2r}}{(2r)!} z^{2r-1}.$$

Integrate with zero constant term and sum $\sum_{j \geq 1} 2^{-2rj} = 1/(4^r - 1)$. This proves (23).

Theorem 4.1 (Partition formulae for the moments). *For every $j, m \geq 0$,*

$$C_j = \sum_{\substack{k_1, \dots, k_j \geq 0 \\ k_1 + 2k_2 + \dots + jk_j = j}} \prod_{r=1}^j \frac{\lambda_r^{k_r}}{k_r!}, \quad (24)$$

$$A_m = \sum_{\substack{k_0, k_1, \dots, k_{\lfloor m/2 \rfloor} \geq 0 \\ k_0 + 2 \sum_{r \geq 1} r k_r = m}} \frac{2^{-k_0}}{k_0!} \prod_{r=1}^{\lfloor m/2 \rfloor} \frac{\beta_r^{k_r}}{k_r!}. \quad (25)$$

At index zero, each sum consists of the unique empty or zero vector and has value one.

Proof. Expand each exponential factor in $M_Y(z) = \prod_{r \geq 1} \exp(\lambda_r z^{2r})$ and collect the coefficient of z^{2j} . Factors with $r > j$ cannot contribute. The second identity follows in the same way from $e^{z/2} \prod_r \exp(\beta_r z^{2r})$. These are the multiplicity-vector versions of the complete Bell-polynomial identities; no recursively defined Bell polynomial is needed. $\square$

Theorem 4.2 (A fully expanded Bernoulli partition value). *Let $n \geq 0$, $0 \leq a \leq 2^n$, and $d = \lfloor n/2 \rfloor$. Then*

$$F(a/2^n) = 2^{-T_n} \sum_{h=0}^{a-1} \varepsilon_h \sum_{\substack{k_0, k_1, \dots, k_d \geq 0 \\ k_0 + 2 \sum_{r=1}^d r k_r = n}} \frac{(2a - 2h - 1)^{k_0}}{k_0!} \prod_{r=1}^d \frac{\lambda_r^{k_r}}{k_r!}. \quad (26)$$

Here every λ_r is the rational expression (22), with B_{2r} explicitly given by (19).

Proof. Substitute (24) into (17), and set $k_0 = n - 2j$. Conversely every vector in (26) determines $j = \sum r k_r$, so the reindexing is bijective. $\square$

4.2 Positive ordered compositions: no Bernoulli numbers

For $m \geq 1$, let $\text{Comp}(m)$ be the finite set of all ordered tuples $\mathbf{r} = (r_1, \dots, r_\ell)$ of positive integers with sum m . Let $\text{Comp}(0)$ contain only the empty tuple. Write

$$S_i(\mathbf{r}) = r_i + \dots + r_\ell. \quad (27)$$

These suffix sums are part of the explicit tuple, not a recursively constructed sequence. There are 2^{m-1} compositions of $m \geq 1$, obtained by choosing cuts among $m - 1$ gaps.

Theorem 4.3 (Positive composition coefficients). *For $m, j \geq 0$,*

$$A_m = \sum_{\mathbf{r} \in \text{Comp}(m)} \prod_{i=1}^{\ell(\mathbf{r})} \frac{1}{(r_i + 1)! (2^{S_i(\mathbf{r})} - 1)}, \quad (28)$$

$$C_j = \sum_{\mathbf{r} \in \text{Comp}(j)} \prod_{i=1}^{\ell(\mathbf{r})} \frac{1}{(2r_i + 1)! (4^{S_i(\mathbf{r})} - 1)}. \quad (29)$$

Every summand is positive.

Proof. First expand the finite products for $\mathbb{E}e^{zX_N}$. A nonconstant contribution selects distinct levels $1 \leq j_1 < \dots < j_\ell \leq N$ and positive exponents $r_1, \dots, r_\ell$ summing to m . Since $\mathbb{E}U^r/r! = 1/(r+1)!$, its coefficient is

$$\prod_{i=1}^{\ell} \frac{2^{-j_i r_i}}{(r_i + 1)!}.$$

The coefficients increase to A_m ; equivalently, dominated convergence applies to X_N^m . Put $j_0 = 0$ and $g_i = j_i - j_{i-1} \geq 1$. Then $\sum_i j_i r_i = \sum_i g_i S_i$, so the unrestricted level sum factors:

$$\sum_{1 \leq j_1 < \dots < j_\ell} 2^{-\sum_i j_i r_i} = \prod_{i=1}^{\ell} \sum_{g_i \geq 1} 2^{-g_i S_i} = \prod_{i=1}^{\ell} \frac{1}{2^{S_i} - 1}. \quad (30)$$

Only finitely many compositions remain, proving (28).

For $Y = \sum_{j \geq 1} 2^{-j} V_j$ with V_j uniform on $[-1, 1]$, odd coordinate moments vanish and $\mathbb{E}V^{2r}/(2r)! = 1/(2r+1)!$. Select positive even exponents $2r_i$. The same gap calculation replaces 2^{S_i} by 4^{S_i} and yields (29). One can first do the argument for a finite product and then pass to the limit; absolute convergence and positivity of its even coefficients justify the passage. $\square$

Theorem 4.4 (A Bernoulli-free closed dyadic sum). *For the same a, n, d as above,*

$$\boxed{F(a/2^n) = 2^{-T_n} \sum_{h=0}^{a-1} \varepsilon_h \sum_{j=0}^d \frac{(2a - 2h - 1)^{n-2j}}{(n-2j)!} \times \sum_{\mathbf{r} \in \text{Comp}(j)} \prod_{i=1}^{\ell(\mathbf{r})} \frac{1}{(2r_i + 1)! (4^{S_i(\mathbf{r})} - 1)}}. \quad (31)$$

Proof. Insert (29) into (17). $\square$

Although the final prefix is signed, its universal coefficient calculation is cancellation-free. In comparison with uncentered compositions, the largest composition weight is only $\lfloor n/2 \rfloor$, not n . The first centered moments are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}, \quad c_4 = \frac{132809}{32531625}. \quad (32)$$

For example, the compositions (2) and (1, 1) give

$$C_2 = \frac{1}{5!(4^2 - 1)} + \frac{1}{3!3!(4^2 - 1)(4 - 1)} = \frac{19}{16200},$$

and multiplication by $4!$ gives c_2 .

4.3 Reciprocal powers and two ways of eliminating the tail

At $a = 1$, (16) gives the particularly simple identity

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}} A_n.} \quad (33)$$

Thus either (25) or (28) is a direct finite formula for these values. For odd depths, a second simplification follows from the bump's half moments:

$$F(2^{-(2j+1)}) = \frac{c_j}{2(2j)! 2^{\binom{2j+1}{2}}}. \quad (34)$$

To prove it without a moment recurrence, note that $\text{up}(t) = \mathbb{P}(X \geq t)$ on $[0, 1]$. Layer-cake integration gives $\mathbb{E}X^{2j+1} = (2j+1) \int_0^1 t^{2j} \text{up}(t) dt = (2j+1)c_j/2$. Now apply (33). This also explains why centered moments can sometimes reduce the work more than raw moments do.

5 A closed formula using only the nonzero binary digits

The prefix in (31) has a terms. For a large numerator this is unnecessarily expensive: the binary expansion can replace it by at most n explicitly known polynomials. The mechanism is the exact Taylor reduction used in the classical arithmetic theory [4]; here its iteration is written as one finite sum and all its coefficients are eliminated.

Lemma 5.1 (One binary step). *For an integer $b \geq 1$ and $0 \leq t \leq 2^{-b}$,*

$$F(2^{-b} + t) + F(t) = 2^{-\binom{b}{2}} \sum_{k=0}^b A_{b-k} \frac{(2^b t)^k}{k!}. \quad (35)$$

Proof. On $[0, 2]$, the bump definition gives $\mathcal{F}(z+2) = -\mathcal{F}(z)$. Therefore the $(b+1)$ st derivative of the left side is zero: by (10), it is a common constant times $\mathcal{F}(2+2^{b+1}t) + \mathcal{F}(2^{b+1}t)$. The left side is consequently a polynomial of degree at most b on the stated interval. At $t = 0$, every derivative of $F(t)$ vanishes, whereas

$$F^{(k)}(2^{-b}) = 2^{k(k+1)/2} F(2^{-(b-k)}) = 2^{bk - \binom{b}{2}} A_{b-k} \quad (0 \leq k \leq b).$$

The endpoint case $k = b$ uses $F(1) = A_0 = 1$. Its finite Taylor polynomial is precisely (35). $\square$

Theorem 5.2 (Explicit binary telescope). *Let $0 < x < 1$ be dyadic, and list its nonzero binary positions as*

$$x = \sum_{i=1}^r 2^{-b_i}, \quad 1 \leq b_1 < \dots < b_r.$$

Define the explicitly known binary suffixes

$$s_i = \sum_{j=i+1}^r 2^{b_i - b_j}, \quad 0 \leq s_i < 1. \quad (36)$$

Then

$$F(x) = \sum_{i=1}^r (-1)^{i-1} 2^{-\binom{b_i}{2}} \sum_{k=0}^{b_i} \frac{s_i^k}{k!} \sum_{\mathbf{r} \in \text{Comp}(b_i-k)} \prod_{j=1}^{\ell(\mathbf{r})} \frac{1}{(r_j+1)! (2^{S_j(\mathbf{r})} - 1)}. \quad (37)$$

The coefficient sum can equivalently be replaced by the finite partition expression (25).

Proof. Write $x_i = \sum_{j=i}^r 2^{-b_j}$ and $x_{r+1} = 0$. Since $x_i = 2^{-b_i} + x_{i+1}$ and $2^{b_i} x_{i+1} = s_i < 1$, the one-step identity says $F(x_i) + F(x_{i+1}) = P_i(s_i)$, with P_i the polynomial in (35). Multiply by $(-1)^{i-1}$ and sum over i . All interior values cancel and $F(0) = 0$. Substitution of (28) gives the displayed formula. This is a proof by a finite telescope, not an instruction to evaluate a recursive sequence. $\square$

For example, $3/8 = 2^{-2} + 2^{-3}$ gives

$$F(3/8) = 2^{-1} (A_2 + \tfrac{1}{2}A_1 + \tfrac{1}{8}A_0) - 2^{-3}A_3 = \frac{73}{288}.$$

After the explicit coefficients through degree n have been computed once, a point with r nonzero bits uses $\sum_i (b_i + 1) \leq r(n+1)$ polynomial terms. This count is not a bit-complexity estimate: large rational numerators and coefficient generation still have costs. Reflection $F(x) = 1 - F(1-x)$ may further reduce the number of useful bits.

6 An integer determinant and an arithmetic certificate

A matrix formula packages the entire evaluation into one determinant with integer entries. It is particularly convenient when a compact exact certificate is preferable to a large expanded composition sum.

Put $d = \lfloor n/2 \rfloor$ and define the integer power sums and row entries

$$S_p(a) = \sum_{h=0}^{a-1} \varepsilon_h (2a - 2h - 1)^p, \quad t_j = \binom{n}{2j} S_{n-2j}(a) \quad (0 \leq j \leq d). \quad (38)$$

Let L_d be the $d \times d$ lower triangular integer matrix with indices $1 \leq k, j \leq d$ and entries

$$(L_d)_{kj} = \begin{cases} (2k+1)(4^k - 1), & j = k, \\ -\binom{2k+1}{2j}, & j < k, \\ 0, & j > k. \end{cases} \quad (39)$$

Finally form the bordered matrix

$$B_{n,a} = \begin{pmatrix} L_d & \mathbf{1} \\ -t_1 \cdots -t_d & t_0 \end{pmatrix}, \quad (40)$$

where $\mathbf{1}$ has d entries all equal to one. When $d = 0$, this means the one-by-one matrix (t_0) .

Theorem 6.1 (Integer determinant formula). *For $n \geq 0$ and $0 \leq a \leq 2^n$,*

$$F(a/2^n) = \frac{\det B_{n,a}}{2^{T_n} n! \prod_{k=1}^d (2k+1)(4^k - 1)}. \quad (41)$$

In particular, the reduced denominator of every value on this grid divides

$$D_n = 2^{T_n} n! (2d+1)!! \prod_{k=1}^d (4^k - 1). \quad (42)$$

No claim is made that D_n is the smallest common denominator.

Proof. The centered MGF obeys $M_Y(2z) = (\sinh z/z)M_Y(z)$, as follows either from the independent uniform decomposition or directly by shifting the product factors. Equating coefficients of z^{2k} , multiplying by $(2k+1)!$, and separating the top term gives

$$(2k+1)(4^k-1)c_k = \sum_{j=0}^{k-1} \binom{2k+1}{2j} c_j.$$

Since $c_0 = 1$, this is the finite system $L_d(c_1, \dots, c_d)^T = \mathbf{1}$. The diagonal entries are nonzero and $\det L_d = \prod_{k=1}^d (2k+1)(4^k-1)$. Subtracting a suitable linear combination of the first d rows from the last row of (40), or using a block determinant, gives

$$\det B_{n,a} = \det L_d \left(t_0 + \sum_{j=1}^d t_j c_j \right).$$

Now use (18). The $d=0$ case follows directly from the same formula. Every entry of $B_{n,a}$ is integral, proving the denominator claim. $\square$

The proof used a moment relation to identify a matrix; the *final expression* does not ask for any moments to be computed. In the literal finite-sum interpretation,

$$\det B_{n,a} = \sum_{\sigma \in S_{d+1}} (-1)^{\#\{(i,j): i < j, \sigma(i) > \sigma(j)\}} \prod_{i=1}^{d+1} (B_{n,a})_{i, \sigma(i)}. \quad (43)$$

Thus even the determinant notation can be removed. An implementation should normally use exact elimination rather than this factorial-size expansion. For $n=2, a=1$, the matrix is

$$B_{2,1} = \begin{pmatrix} 9 & 1 \\ -1 & 1 \end{pmatrix}, \quad F(1/4) = \frac{10}{2^3 2! 9} = \frac{5}{72}.$$

7 Centered splines and exact cutoff polynomials

This section explains why a short moment-free formula exists. It is important that the cutoff dependence terminates *exactly*, not just to a prescribed asymptotic accuracy.

7.1 The finite centered CDF

For $N \geq 1$, set $h = 2^{-N}$ and define

$$H_N(x) = \mathbb{P}(X_N + h/2 \leq x). \quad (44)$$

The finite random variable is centered at $1/2$, just like X . Its CDF is a piecewise polynomial of degree N , with possible knots $h(k+1/2)$, $0 \leq k < 2^N$. For a dyadic argument $x = a/2^n$ and $N \geq n \geq 1$, write $M = a2^{N-n}$. The box formula immediately gives

$$H_N(a/2^n) = \frac{1}{N! 2^{N(N+1)/2}} \sum_{k=0}^{M-1} \varepsilon_k (2M - 2k - 1)^N. \quad (45)$$

The sum is empty at $a=0$. At $a=2^n$ it evaluates to one. Although the density knots are numerous, the required threshold is always halfway between two consecutive possible knots.

7.2 A finite-jet deconvolution lemma

The reciprocal centered MGF has the local series

$$\frac{1}{M_Y(z)} = \sum_{j \geq 0} R_j z^{2j}, \quad R_j = \sum_{\substack{k_1, \dots, k_j \geq 0 \\ \sum_{r=1}^j r k_r = j}} \prod_{r=1}^j \frac{(-\lambda_r)^{k_r}}{k_r!}. \quad (46)$$

Here $R_0 = 1$, and λ_r is the explicit number in (22). This coefficient formula follows by expanding $\exp(-\sum_r \lambda_r z^{2r})$; it requires no reciprocal-series recurrence.

Lemma 7.1 (Local polynomial smoothing is invertible on a finite jet). *Suppose a piecewise polynomial H agrees with a polynomial $p(s)$ of degree at most N at the points $x + s$, $|s| \leq \eta$. Let Z be a symmetric random variable supported on $[-1, 1]$ with a smooth density, and suppose $f(t) = \mathbb{E}H(t - \eta Z)$. Write $M_Z(z) = \mathbb{E}e^{zZ}$ and $M_Z(z)^{-1} = \sum_{j \geq 0} \rho_j z^{2j}$. If the derivatives through order N of the convolution are obtained by convolving the corresponding piecewise derivatives of H , then*

$$H(x) = \sum_{j=0}^{\lfloor N/2 \rfloor} \rho_j \eta^{2j} f^{(2j)}(x). \quad (47)$$

For the CDF splines used here, these derivative identities hold.

Proof. Let D denote differentiation of polynomials in s and let $J_N f(s) = \sum_{r=0}^N f^{(r)}(s) s^r / r!$. Differentiating the convolution as stipulated, and using the polynomial valid across the whole averaging interval, shows coefficient by coefficient that

$$J_N f = M_Z(-\eta D)p = M_Z(\eta D)p.$$

This is an identity of polynomials, with the differential operator truncated at order N . The second equality uses symmetry. Since $D^{N+1} = 0$ on this space and $M_Z(0) = 1$, its inverse is the finite operator obtained by truncating $1/M_Z(\eta D)$. Apply the inverse and evaluate at $s = 0$. This proves (47) without a Taylor-series convergence assumption on f .

For an N -uniform CDF, derivatives through order $N - 1$ are continuous; the N th derivative is a bounded piecewise constant function. These are also its distributional derivatives through order N . Convolution with a smooth density therefore gives the derivative identities. Values at the finitely many knot endpoints do not affect any integral. The same argument works for a density spline, with its degree in place of N . $\square$

Theorem 7.2 (Exact polynomial in the cutoff scale). *Fix $n \geq 1$, $0 \leq a \leq 2^n$, and $x = a/2^n$. For every $N \geq n$,*

$$\boxed{H_N(x) = \sum_{j=0}^d R_j 2^{-2j(N+1)} F^{(2j)}(x), \quad d = \lfloor n/2 \rfloor.} \quad (48)$$

Thus there is one polynomial $P_x(z)$ of degree at most d , independent of N , such that

$$H_N(x) = P_x(4^{-N}), \quad P_x(0) = F(x). \quad (49)$$

Proof. The head–tail decomposition takes the centered form

$$X \stackrel{d}{=} X_N + h/2 + (h/2)Y', \quad F(t) = \mathbb{E}H_N(t - (h/2)Y'),$$

where Y' is an independent copy of Y . Since $x = Mh$, the interval $[x - h/2, x + h/2]$ contains no knot in its interior. On this entire interval $H_N(x + s)$ is represented by one polynomial of degree

N ; continuity includes its endpoints. Apply Lemma 7.1 with $\eta = h/2$ and $Z = Y'$. By Lemma 2.1, all derivatives of F at x above order n vanish. This removes all even terms with $2j > n$ and proves (48). The endpoints $x = 0, 1$ can either be treated by the same flatness statement or observed directly: both CDFs there are 0, 1. The polynomial in (49) is explicitly $P_x(z) = \sum_{j=0}^d R_j 4^{-j} F^{(2j)}(x) z^j$. $\square$

Only the constant coefficient of P_x is needed. Its other coefficients can be entirely eliminated by interpolation. For understanding the identity, the first reciprocal coefficients are useful:

$$R_0 = 1, \quad R_1 = -\frac{1}{18}, \quad R_2 = \frac{31}{16200}, \quad R_3 = -\frac{2347}{42865200}. \quad (50)$$

For instance, $F''(1/4) = 8$ and higher even derivatives vanish, giving the particularly transparent exact formula

$$H_N(1/4) = \frac{5}{72} - \frac{1}{9} 4^{-N} \quad (N \geq 2). \quad (51)$$

7.3 The degree bound is sharp at reduced dyadic arguments

Proposition 7.3 (Nonzero reciprocal coefficients and exact degree). *For every $j \geq 0$, $(-1)^j R_j > 0$. If $0 < a < 2^n$, a is odd, and $n \geq 1$, the polynomial $P_{a/2^n}$ in (49) has degree exactly $\lfloor n/2 \rfloor$.*

Proof. The classical product for $\sinh z/z$ [9, §4.36] gives, in a neighborhood of the origin,

$$\frac{1}{M_Y(z)} = \prod_{\nu \geq 1} \prod_{m \geq 1} \left(1 + \frac{z^2}{4^\nu \pi^2 m^2} \right)^{-1}. \quad (52)$$

The positive numbers $\alpha_{\nu,m} = 1/(4^\nu \pi^2 m^2)$ have a finite sum. Hence the absolute value of the coefficient of z^{2j} is the convergent strictly positive complete homogeneous sum of degree j in this summable alphabet, and its sign is $(-1)^j$. In particular it is nonzero. This infinite product is used only to prove a sign; the evaluation formula for R_j remains the finite expression (46).

If $n = 2d$, then $F^{(2d)}(a/2^n) = 2^{n(n+1)/2} \mathcal{F}(a)$ is nonzero because a is odd. If $n = 2d + 1$ and $d \geq 1$, the relevant signed value is $\mathcal{F}(a/2)$, whose magnitude is $F(1/2) = 1/2$ by the bump decomposition. It too is nonzero. For $n = 1$, the constant polynomial has value $1/2$. Combining these facts with $R_d \neq 0$ proves the assertion. $\square$

Reduction of the dyadic fraction is therefore useful: an unreduced depth still gives a correct upper bound and a correct formula, but may introduce unnecessary interpolation nodes.

8 A moment-free quarter-base formula

8.1 The geometric interpolation row

For $0 < q < 1$ and $0 \leq j \leq d$, define

$$w_{d,j}(q) = \prod_{\substack{0 \leq \ell \leq d \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} \quad (53)$$

$$= \frac{(-1)^{d-j} q^{(d-j)(d-j+1)/2}}{(q; q)_j (q; q)_{d-j}} = \frac{(-1)^{d-j} q^{\binom{d-j+1}{2}}}{(q; q)_d} \left[\begin{matrix} d \\ j \end{matrix} \right]_q. \quad (54)$$

Lemma 8.1 (Constant extraction on a geometric grid). *For every polynomial P of degree at most d and every nonzero z ,*

$$P(0) = \sum_{j=0}^d w_{d,j}(q) P(zq^j). \quad (55)$$

Equivalently,

$$\sum_{j=0}^d w_{d,j}(q) q^{jr} = \begin{cases} 1, & r = 0, \\ 0, & 1 \leq r \leq d. \end{cases} \quad (56)$$

The generating polynomial of this row is

$$\sum_{j=0}^d w_{d,j}(q) v^j = \prod_{r=1}^d \frac{v - q^r}{1 - q^r}. \quad (57)$$

Proof. The Lagrange basis polynomial for the node zq^j , evaluated at zero, is exactly (53); the common factor z cancels. This proves the extraction identity and its monomial specializations. Separating the factors with $\ell < j$ and $\ell > j$ in the product gives (54). The generating polynomial has zeros at $q, \dots, q^d$ by (56), and value one at $v = 1$, proving (57). These identities also follow from the finite q -binomial theorem [8, §17.2], but no such theorem is needed for this proof. $\square$

Theorem 8.2 (Quarter-base, moment-free dyadic evaluation). *Let $n \geq 1$, $0 \leq a \leq 2^n$, and $d = \lfloor n/2 \rfloor$. Then*

$$F(a/2^n) = \sum_{j=0}^d w_{d,j}(1/4) H_{n+j}(a/2^n). \quad (58)$$

Eliminating the splines gives the completely explicit double finite sum

$$F(a/2^n) = \sum_{j=0}^d \frac{(-1)^{d-j} 4^{-\binom{d-j+1}{2}}}{(1/4; 1/4)_j (1/4; 1/4)_{d-j} (n+j)! 2^{(n+j)(n+j+1)/2}} \times \sum_{k=0}^{2^j a - 1} (-1)^{s_2(k)} (2^{j+1} a - 2k - 1)^{n+j}. \quad (59)$$

No moments, Bernoulli numbers, derivative values, recursively defined coefficients, limits, or coefficient extractors occur in this formula.

Proof. Apply Lemma 8.1 to the degree- d polynomial P_x from theorem 7.2, with $z = 4^{-n}$ and $q = 1/4$. Then substitute (45) at $N = n + j$. The case $d = 0$ uses the single weight one and needs no separate interpolation argument. $\square$

The first rows, useful for direct implementation, are

$$\begin{aligned} d = 0 : & \quad (1), \\ d = 1 : & \quad \frac{1}{3}(-1, 4), \\ d = 2 : & \quad \frac{1}{45}(1, -20, 64), \\ d = 3 : & \quad \frac{1}{2835}(-1, 84, -1344, 4096). \end{aligned}$$

The largest spline level required is $n + \lfloor n/2 \rfloor$. The corresponding half-base construction has largest level $2n$. This is a reduction in the number and size of finite splines, not a claim that the method is faster than every other exact evaluator.

8.2 Arbitrary starting depths and arbitrary nodes

Consecutive levels need not start at n . For every $N_0 \geq n$,

$$F(a/2^n) = \sum_{j=0}^d w_{d,j}(1/4) H_{N_0+j}(a/2^n). \quad (60)$$

More generally, for any distinct integers $N_0, \dots, N_d \geq n$,

$$F(a/2^n) = \sum_{j=0}^d \left(\prod_{\substack{0 \leq \ell \leq d \\ \ell \neq j}} \frac{-4^{-N_\ell}}{4^{-N_j} - 4^{-N_\ell}} \right) H_{N_j}(a/2^n). \quad (61)$$

These are immediate Lagrange evaluations of the same polynomial. They permit reuse of already available spline values. The degree bound also explains the universal node count: with $r \leq d$ nonzero distinct nodes, the polynomial $\prod_{i=1}^r (z - z_i)$ vanishes at every sampled node but not at zero. Consequently no rule using only those samples can recover $P(0)$ for *every* polynomial of degree at most d . This is a statement about universal polynomial recovery, not a lower bound for all special-function algorithms.

8.3 Size and signs of the interpolation coefficients

The sign of $w_{d,j}(q)$ is $(-1)^{d-j}$. Substituting $v = -1$ into (57) therefore proves

$$\sum_{j=0}^d |w_{d,j}(q)| = \prod_{r=1}^d \frac{1+q^r}{1-q^r} \leq \prod_{r=1}^{\infty} \frac{1+q^r}{1-q^r} < \infty. \quad (62)$$

The infinite product is finite because its logarithm is bounded by a constant times $\sum_r q^r$. For $q = 1/4$, the limiting amplification is approximately 1.96926, whereas for $q = 1/2$ it is approximately 8.25599. Thus the interpolation step itself does not become progressively ill-conditioned in the absolute maximum norm. Nevertheless, the alternating power sum inside an individual spline can have severe cancellation. Exact integer and rational arithmetic is the appropriate default. A nonnegative interpolation row is impossible when $d \geq 1$: positive nodes and nonnegative weights summing to one cannot have zero first moment.

9 How the half-base formula fits into the picture

The quarter-base formula is not obtained by replacing $1/2$ by $1/4$ in an existing identity. The centering removes all odd cutoff powers and changes the required interpolation degree. A shift parameter makes the distinction precise.

For $0 \leq \alpha \leq 1$, set

$$H_{N,\alpha}(x) = \mathbb{P}(X_N + \alpha 2^{-N} \leq x). \quad (63)$$

For $x = a/2^n$, $N \geq n \geq 1$, and $M = a2^{N-n}$,

$$H_{N,\alpha}(a/2^n) = \frac{2^{-\binom{N}{2}}}{N!} \sum_{k=0}^{M-1} \varepsilon_k (M - k - \alpha)^N. \quad (64)$$

At $\alpha = 0$ or 1 the extra boundary terms have zero N th power, so this same finite expression includes both endpoints. For arbitrary complex α , define $H_{N,\alpha}$ at the displayed dyadic argument by the polynomial on the right; it no longer has a probability interpretation, but remains a finite algebraic object.

Proposition 9.1 (General shift and half-base extraction). *For $n \geq 1$, $0 \leq a \leq 2^n$, and every $\alpha \in \mathbb{C}$,*

$$F(a/2^n) = \sum_{j=0}^n w_{n,j}(1/2) H_{n+j,\alpha}(a/2^n). \quad (65)$$

Equivalently,

$$\boxed{F(a/2^n) = \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{j=0}^n \frac{[n]_{1/2}}{2^{j(j-1)}(n+j)!} \times \sum_{k=0}^{2^j a - 1} \varepsilon_k(k - 2^j a + \alpha)^{n+j}.} \quad (66)$$

Proof. First suppose $0 \leq \alpha \leq 1$ and put $h = 2^{-N}$. The head–tail identity reads $F(t) = \mathbb{E}H_{N,\alpha}(t - h(X' - \alpha))$. At $x = Mh$, the entire averaging interval $[x - h(1 - \alpha), x + h\alpha]$ lies in one polynomial cell of $H_{N,\alpha}$. Its endpoints are consecutive possible knots. Repeat the finite-jet proof of Lemma 7.1, without symmetry. Writing $Z = X' - \alpha$, the inverse operator is $1/M_Z(-hD)$, and therefore

$$H_{N,\alpha}(x) = \sum_{r=0}^n \gamma_r(\alpha) h^r F^{(r)}(x), \quad \gamma_r(\alpha) = [z^r] \frac{1}{M_Z(-z)}.$$

Here $\gamma_0 = 1$ and the coefficients do not depend on N . The formula is an exact polynomial in 2^{-N} of degree at most n . Interpolation at $n + 1$ consecutive depths proves (65). Both sides, when the finite formula (64) is substituted, are polynomials in α . Equality throughout $[0, 1]$ implies equality for all complex α .

Insert the explicit weights into (65) and replace $(M - k - \alpha)^{n+j}$ by $(-1)^{n+j}(k - M + \alpha)^{n+j}$. This cancels the weight's sign $(-1)^{n-j}$. The powers of two combine as

$$\binom{n-j+1}{2} + \binom{n+j}{2} = n^2 + j(j-1),$$

which gives (66). $\square$

Formula (66) is the unit-interval form of the expression posed by Reshetnikov in 2019 [6], subsequently developed in the repository and its companion exposition [1, 2]. The repository also treats the signed extension directly. Our folding rule (7) is sufficient to evaluate it everywhere without reproving that unrestricted-numerator identity. At $\alpha = 1/2$, $Z = Y'/2$ is symmetric and the odd coefficients vanish. The degree drops from at most n in 2^{-N} to at most $\lfloor n/2 \rfloor$ in 4^{-N} . This is precisely the passage from (65) to (58).

10 Direct Rvachev formulae, derivatives, and primitives

10.1 A finite density instead of a finite CDF

Although (6) already determines every bump value, a direct formula is useful when the available data are finite sinc products. Let

$$Y_N = \sum_{j=1}^N 2^{-j} V_j, \quad V_j \sim \text{Unif}[-1, 1], \quad u_N = \text{density of } Y_N. \quad (67)$$

With the convention $\hat{f}(t) = \int f(x) e^{-2\pi i t x} dx$,

$$\hat{u}_N(t) = \prod_{j=0}^{N-1} \text{sinc}(\pi t / 2^j), \quad \text{sinc}(z) = \frac{\sin z}{z}. \quad (68)$$

Thus an approximation indexed by the last factor $j = m$ is u_{m+1} in our notation. This offset matters in exact cutoff formulae.

For $|a| < 2^n$ and $N > n \geq 0$, put

$$A = 2^N + a2^{N-n}. \quad (69)$$

This is a positive even integer. The density version of the box formula is

$$u_N(a/2^n) = \frac{1}{(N-1)! 2^{N(N-1)/2}} \sum_{k=0}^{A/2-1} \varepsilon_k (A-1-2k)^{N-1}. \quad (70)$$

Indeed, $Y_N = \sum_{j=1}^N 2^{1-j} U_j - (1 - 2^{-N})$. Differentiate the weighted-uniform CDF once; its subset thresholds are $2^{-N}(2k - 2^N + 1)$. At the specified argument, the positive terms have $2k < A - 1$, giving precisely the displayed range and normalization.

Theorem 10.1 (Direct quarter-base extraction for the bump). *Let $n \geq 0$, $|a| < 2^n$, and $d = \lfloor n/2 \rfloor$. Then*

$$u_N(a/2^n) = \sum_{j=0}^d R_j 4^{-Nj} \text{up}^{(2j)}(a/2^n) \quad (N > n), \quad (71)$$

$$\text{up}(a/2^n) = \sum_{j=0}^d w_{d,j}(1/4) u_{n+1+j}(a/2^n). \quad (72)$$

Each u_{n+1+j} is the finite integer power sum (70). For $|a| \geq 2^n$, the bump value is zero.

Proof. Now $Y \stackrel{d}{=} Y_N + hY'$ with $h = 2^{-N}$, so $\text{up}(t) = \mathbb{E}u_N(t - hY')$. The knots of u_N are odd multiples of h . Since $N > n$, the number $x/h = a2^{N-n}$ is even; hence $[x-h, x+h]$ is one polynomial cell. On that cell u_N has degree $N-1$. Use the finite-jet inverse lemma with this degree and $\eta = h$. All derivatives of up above order n vanish by Lemma 2.1, giving (71). Constant extraction with $q = 1/4$ yields (72). The case $N = 1, n = 0, a = 0$ is the constant density on its interior and follows by the same degree-zero argument. Outside $[-1, 1]$, and at its endpoints, the compactly supported smooth bump is zero. $\square$

At $x = 1/4$, the direct density construction yields

$$u_3(1/4) = \frac{15}{16}, \quad u_4(1/4) = \frac{179}{192}, \quad \text{up}(1/4) = \frac{4u_4(1/4) - u_3(1/4)}{3} = \frac{67}{72}.$$

In fact $\text{up}''(1/4) = -8$, so the whole tail of the sequence satisfies

$$u_N(1/4) = \frac{67}{72} + \frac{4}{9} 4^{-N} \quad (N \geq 3). \quad (73)$$

Unlike numerical extrapolation from an asymptotic expansion, two suitable consecutive values determine the limit here with zero remainder.

10.2 All derivatives at dyadics reduce to values

The preceding formulae also give closed expressions for every derivative. For an interior dyadic $x = a/2^n$ and $0 \leq r \leq n$, use

$$F^{(r)}(x) = 2^{r(r+1)/2} \mathcal{F}(2^r x), \quad (74)$$

$$\text{up}^{(r)}(x) = 2^{r(r+1)/2} \mathcal{F}(2^r(x+1)). \quad (75)$$

In the first line, $x \in (0, 1)$; in the second, $x \in (-1, 1)$. Fold the signed argument using (7), then evaluate its unit-interval value by any finite formula above. The denominator depth is at most $n - r$. All orders $r > n$ give zero. At F 's boundary points the positive-order derivatives are zero; at the bump's boundary points all orders, including zero, vanish. This procedure consists of finite substitutions, not a chain of function-value recurrences. It can be substituted directly into (48) or (71) to make every polynomial coefficient rational and explicit.

10.3 Iterated primitives

For an integer $r \geq 1$, repeated integration has an equally simple exact form:

$$\frac{1}{(r-1)!} \int_0^x (x-t)^{r-1} \mathcal{F}(t) dt = 2^{\binom{r}{2}} \mathcal{F}(x/2^r) \quad (x \geq 0), \quad (76)$$

$$\frac{1}{(r-1)!} \int_0^x (x-t)^{r-1} F(t) dt = 2^{\binom{r}{2}} F(x/2^r) \quad (0 \leq x \leq 1), \quad (77)$$

$$\frac{1}{(r-1)!} \int_{-1}^x (x-t)^{r-1} \text{up}(t) dt = 2^{\binom{r}{2}} F((x+1)/2^r) \quad (-1 \leq x \leq 1). \quad (78)$$

For (76), differentiate the right side r times and use (10); the total power of two is $\binom{r}{2} - r^2 + r(r+1)/2 = 0$. Its first r initial data at zero vanish, so repeated FTC proves the identity. The second line restricts the first to the unit interval. The third uses $\text{up}(t) = \mathcal{F}(t+1)$ and translates the lower endpoint; the final argument belongs to $[0, 1]$. Thus these integrals at dyadic endpoints are rational and have exactly the same recurrence-free evaluations at a larger denominator depth.

11 The reciprocal-integer-base extension

The same mechanism is not confined to base two. The dyadic case has the extra convenience that all subset sums fill one consecutive integer block, allowing a single Thue–Morse prefix. Other integer bases require a Boolean cube sum instead.

11.1 The normalized base- b law

Let $b \geq 2$ be an integer and define

$$X_b = (b-1) \sum_{j \geq 1} b^{-j} U_j, \quad F_b(x) = \mathbb{P}(X_b \leq x). \quad (79)$$

It has support $[0, 1]$, reflection symmetry, and a smooth density, by the same absolutely convergent uniform-series and Fourier arguments used above. In particular, it is flat at both boundary points. The raw moment coefficients have the positive finite formula

$$A_m^{(b)} = \frac{\mathbb{E} X_b^m}{m!} = (b-1)^m \sum_{\mathbf{r} \in \text{Comp}(m)} \prod_{i=1}^{\ell(\mathbf{r})} \frac{1}{(r_i+1)! (b^{S_i(\mathbf{r})} - 1)}. \quad (80)$$

The proof repeats the gap summation (30), replacing 2 by b and including the common scale $(b-1)^m$.

For $n \geq 1$, $0 \leq a \leq b^n$, and $\delta = (\delta_0, \dots, \delta_{n-1}) \in \{0, 1\}^n$, put

$$\Delta_\delta = a - (b-1) \sum_{j=0}^{n-1} \delta_j b^j. \quad (81)$$

Theorem 11.1 (Finite base- b arithmetic). *For these parameters,*

$$F_b(a/b^n) = \frac{b^{-\binom{n}{2}}}{(b-1)^n} \sum_{\substack{\delta \in \{0,1\}^n \\ \Delta_\delta \geq 1}} (-1)^{|\delta|} \sum_{m=0}^n A_m^{(b)} \frac{(\Delta_\delta - 1)^{n-m}}{(n-m)!}. \quad (82)$$

Together with (80), this is a finite rational expression. Consequently F_b takes rational values at every base- b rational argument.

Proof. Split X_b into its first n coordinates and the independent tail $b^{-n}X'_b$. Apply the finite box formula to the first coordinates, whose weights have product $(b-1)^n b^{-n(n+1)/2}$. Conditional on the tail, the truncated power is $b^{-n^2}(\Delta_\delta - X'_b)_+^n$. Since Δ_δ is integral and $X'_b \in [0, 1]$, it is zero for $\Delta_\delta \leq 0$ and is an ordinary polynomial for $\Delta_\delta \geq 1$. Reflection replaces $1 - X'_b$ by X'_b , so expansion of $(\Delta_\delta - 1 + X'_b)^n/n!$ gives the inner sum. The scalar factors combine to the one displayed. $\square$

11.2 Parity-reduced extraction at base b^{-2}

For $N \geq 1$, set

$$X_{b,N} = (b-1) \sum_{j=1}^N b^{-j} U_j, \quad \mathbf{H}_{N,b}(x) = \mathbb{P}(X_{b,N} + b^{-N}/2 \leq x).$$

For $N \geq n$, let $M = ab^{N-n}$. Its direct finite evaluation is

$$\begin{aligned} \mathbf{H}_{N,b}(a/b^n) &= \frac{1}{2^N (b-1)^N b^{\binom{N}{2}} N!} \\ &\times \sum_{\delta \in \{0,1\}^N} (-1)^{|\delta|} \left(2M - 1 - 2(b-1) \sum_{j=0}^{N-1} \delta_j b^j \right)_+^N. \end{aligned} \quad (83)$$

This is still a finite rational formula; positive part here is just a comparison of explicitly given integers.

Theorem 11.2 (Moment-free integer-base extraction). *For $n \geq 1$, $0 \leq a \leq b^n$, and $d = \lfloor n/2 \rfloor$,*

$$F_b(a/b^n) = \sum_{j=0}^d w_{d,j}(b^{-2}) \mathbf{H}_{n+j,b}(a/b^n). \quad (84)$$

All quantities on the right are finite products or the Boolean cube sum (83).

Proof. The self-similarity $X_b \stackrel{d}{=} ((b-1)U + X'_b)/b$ gives

$$F'_b(x) = \frac{b}{b-1} \{F_b(bx) - F_b(bx - (b-1))\}. \quad (85)$$

Iterating this differentiation identity n times writes $F_b^{(n)}(x)$ as a finite linear combination of values $F_b(b^n x - k)$ with integral k . Further differentiation and evaluation at $x = a/b^n$ involves only positive-order derivatives of F_b at integers. All such derivatives vanish: integers are boundary points or lie outside the support of the density. Thus $F_b^{(r)}(a/b^n) = 0$ for $r > n$.

Write $Z_b = 2X_b - 1$, a symmetric variable supported on $[-1, 1]$, and $h = b^{-N}$. The head-tail identity is $F_b(t) = \mathbb{E} \mathbf{H}_{N,b}(t - (h/2)Z'_b)$. Every knot of the centered finite CDF is an odd half-integer multiple of h , while a/b^n is an integer multiple of h for $N \geq n$. The averaging interval therefore lies in one polynomial cell; missing knots cause no problem. The finite-jet inverse lemma proves that $\mathbf{H}_{N,b}(a/b^n)$ is an exact polynomial in b^{-2N} of degree at most d , with constant coefficient $F_b(a/b^n)$. Apply geometric constant extraction with $q = b^{-2}$. $\square$

For example, the explicit sums give

$$F_3(1/9) = \frac{7}{576}, \quad F_3(1/27) = \frac{1}{6912}, \quad F_4(1/16) = \frac{1}{240}, \quad F_4(1/64) = \frac{1}{51840}.$$

At $b = 2$, the cube sums collapse to the dyadic prefix and all preceding formulae are recovered. For $b > 2$, some of the highest jets can vanish for additional geometric reasons; we assert the degree *upper bound*, not the dyadic exact-degree statement. In particular, this extension does not assert that the base- b functions have all the same local analytic properties as the dyadic Fabius function.

12 Worked exact values and reproducible verification

12.1 Small closed-form examples

The first quarter-base row already gives

$$F(1/4) = \frac{4(13/192) - 1/16}{3} = \frac{5}{72}, \quad (86)$$

where $H_2(1/4) = 1/16$ and $H_3(1/4) = 13/192$. For a denominator of eight, the same two weights apply:

$$H_3(3/8) = \frac{97}{384}, \quad H_4(3/8) = \frac{389}{1536}, \quad F(3/8) = \frac{4H_4(3/8) - H_3(3/8)}{3} = \frac{73}{288}.$$

At depth four, three levels suffice:

$$H_4(1/16) = \frac{1}{24576}, \quad H_5(1/16) = \frac{121}{1966080}, \quad H_6(1/16) = \frac{6331}{94371840}.$$

$$F(1/16) = \frac{H_4(1/16) - 20H_5(1/16) + 64H_6(1/16)}{45} = \frac{143}{2073600}.$$

Each spline fraction comes from a finite signed sum of odd integer powers; no previously known value of F was inserted into the extraction.

a	$F(a/16)$	a	$F(a/16)$
1	143/2073600	5	305857/2073600
2	1/288	6	73/288
3	46657/2073600	7	777743/2073600
4	5/72	8	1/2

Table 2: Exact dyadic values. The remaining interior entries follow by $F((16 - a)/16) = 1 - F(a/16)$. The bump values follow from $\text{up}(a/16) = F((16 - |a|)/16)$.

12.2 What the accompanying program verifies

The file `verify.py` uses only the Python standard library and `fractions.Fraction`. It implements five separate evaluations: the centered composition formula, the centered partition formula, the binary telescope, the bordered determinant, and the quarter-base spline formula. The determinant is evaluated by exact rational elimination. Composition sets are enumerated by cut masks, and partition multiplicities by a bounded Cartesian product; neither coefficient generator uses a moment recurrence. Memoization merely avoids repeating an explicit sum.

The completed verification run recorded in `verification.json` contains the following counts.

Check	Cases	Range or description
Centered coefficient equality	9	C_j from compositions and partitions, $0 \leq j \leq 8$
Five dyadic value formulae	520	Every $0 \leq a \leq 2^n$, $0 \leq n \leq 8$; reflection and refinement are checked too
Exact cutoff polynomial	360	Interior points at $1 \leq n \leq 6$, with $N = n, n+1, n+3$
Direct bump extraction	132	Every $-2^n \leq a \leq 2^n$, $0 \leq n \leq 5$
Half-base shift independence	65	Selected points through depth five, $\alpha = 0, 1/2, 1, -2, 2/3$
Other integer bases	1252	Every $0 \leq a \leq b^n$ for $b = 3, 4, 5$ and $1 \leq n \leq 4$

The 520 grid cases are numerator–depth pairs, not 520 distinct rational arguments: unreduced representations are intentionally included. Eight standard reciprocal-power values are checked separately as anchors. All comparisons are exact equality of rational numbers; there is no floating-point tolerance. These tests check the implementations and indexing. They are not a replacement for the proofs, nor are they a claim that the new presentation has been formalized in Lean.

To reproduce the report, run

```
python verify.py --json verification.json
```

The default maximum dyadic depth is eight. A larger value can be requested with `–max-depth`; the full-grid tests grow exponentially. The other test ranges remain fixed.

12.3 A minimal quarter-base evaluator

The following self-contained version shows how little algebra is required for the moment-free formula. The full verification program supplies the other methods and cross-checks.

```
from fractions import Fraction
from math import factorial

def fabius_quarter(a: int, n: int) -> Fraction:
    """Exact bounded Fabius value at a / 2**n."""
    if not isinstance(a, int) or not isinstance(n, int):
        raise TypeError("a and n must be integers")
    if n < 0 or not 0 <= a <= 2**n:
        raise ValueError("require n >= 0 and 0 <= a <= 2**n")
    if a == 0 or a == 2**n:
        return Fraction(int(a != 0))
    # Reduce the denominator before choosing the number of levels.
    while n > 0 and a % 2 == 0:
        a //= 2
        n -= 1
    d = n // 2
    q = Fraction(1, 4)
    result = Fraction(0)
    for j in range(d + 1):
        weight = Fraction(1)
        for ell in range(d + 1):
            if ell != j:
                weight *= -q**ell / (q**j - q**ell)
        N = n + j
        M = a * 2**j
        numerator = sum(
            (-1 if bin(k).count("1") % 2 else 1)
            * (2*M - 2*k - 1)**N
            for k in range(M)
        )
        spline = Fraction(numerator,
                          factorial(N) * 2**(N*(N+1)//2))
        result += weight * spline
    return result
```

This evaluator makes no call to a function-value or moment recurrence. The finite loops implement the two sums and the weight product literally. Reduction of the input is optional mathematically, but useful computationally.

13 Choosing a representation and possible extensions

13.1 Which finite formula to use

For a single value with a sparse binary expansion, the binary telescope usually avoids the largest prefix sums. For many values on one grid, the centered moment formula permits the same explicitly computed C_j to be reused. The composition expression is particularly transparent when positive coefficient terms and elementary denominators are desirable. The partition expression connects directly to cumulants and is well suited to symbolic manipulation. The determinant is a compact integer certificate, with a visible common denominator. The quarter-base construction is the most direct answer when *no auxiliary arithmetic coefficients at all* are desired: it contains only a short geometric interpolation row and finite power sums.

These are algebraically equivalent descriptions, but they are not identical algorithms. The number of summands does not alone measure performance. Large powers and cancellation, bit lengths of rational intermediates, coefficient reuse, and whether one value or a whole grid is required all matter. The exact arithmetic in the program is intentionally straightforward; it is a verification reference rather than a claim of optimal implementation.

13.2 Further deductions that do not require additional conjectures

A finite subset of the available cutoff coefficients can vanish at special points. Whenever the nonzero orders are known, one may interpolate only those monomials rather than the complete degree space. The exact-degree result at reduced dyadics shows that the highest allowed order does not vanish, but does not exclude gaps at lower orders. This suggests point-adapted extraction rows, with their validity proved by the corresponding finite moment equations.

The local inverse proof applies more generally whenever a smooth law is an independent sum of a finite spline head and a scaled tail, the support of the tail fits in one polynomial cell at the evaluation point, and the relevant jet of the limiting function terminates. Symmetry removes the odd cutoff powers; other coefficient zeros can remove further powers. One can therefore separate the search for additional exact formulae into a geometric question about knot cells and an algebraic question about the nonzero jet orders. The integer-base theorem is one concrete realization of this separation.

13.3 Conclusion

The requested rational values admit several closed forms in the strict finite sense. The hidden moment recurrences in the classical formula can be eliminated by explicit multiplicity partitions, by positive ordered compositions, or by an integer determinant. Binary digits provide a finite telescope with no long signed prefix. Most importantly, centered finite convolutions have an exact polynomial dependence on the squared tail scale at dyadic arguments. Removing that polynomial by geometric interpolation gives (59), a moment-free $q = 1/4$ formula requiring only $\lfloor n/2 \rfloor + 1$ spline levels. Reflection and folding cover all Rvachev and signed Fabius values, and the same finite algebra extends to all derivatives, normalized iterated primitives, and reciprocal-integer-base uniform laws.

A A direct finite definition of every index set

To remove any ambiguity about the word “finite”, all auxiliary indexing can be reduced to bounded integer tuples. A composition of $m \geq 1$ is specified by a length $1 \leq \ell \leq m$ and a tuple $(r_1, \dots, r_\ell) \in \{1, \dots, m\}^\ell$ with $r_1 + \dots + r_\ell = m$. The empty tuple is the only composition of zero. A partition multiplicity vector of weight j lies in

$$\prod_{r=1}^j \{0, 1, \dots, \lfloor j/r \rfloor\} \quad \text{and satisfies} \quad \sum_{r=1}^j r k_r = j.$$

In a raw partition of weight m , add $k_0 \in \{0, \dots, m\}$ and impose $k_0 + 2 \sum_r r k_r = m$. All binomial coefficients can be read as factorial quotients in their stated ranges. All Bernoulli numbers are the finite power sums (19). Every binary sign is the explicit digit product following (1). Every determinant is the permutation sum (43). Every q -Pochhammer symbol in a final value formula is the finite product (2). There is therefore no unstated recursive sequence anywhere in the evaluation layer.

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